PRACTICAL FILE

OOPs using C++

Object Oriented Programming

B.Tech CSE • 3rd Semester

SUBMITTED BY

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B

SUBMITTED TO

FACULTY
Amanpreet Maam

SUBJECT
OOPs using C++

COURSE
B.Tech CSE

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Date

Signature of Amanpreet Maam

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Q1. Write a C++ program to print Hello Amanpreet Maam.

Program:

```
#include <iostream>
using namespace std;
int main() {
    cout << "Hello Amanpreet Maam" << endl;
    return 0;
}
```

Output:



Q2. Introduce yourself in 10 lines using a C++ program.

Program:

```
#include <iostream>
using namespace std;
int main() {
    cout << "1. Name     : Mausham Kumar Chaudhary" << endl;
    cout << "2. Roll No  : AN125309180085" << endl;
    cout << "3. Course   : B.Tech CSE" << endl;
    cout << "4. Semester : 3rd" << endl;
    cout << "5. Section  : B" << endl;
    cout << "6. College  : Guru Kashi University" << endl;
    cout << "7. Subject  : OOPs using C++" << endl;
    cout << "8. Faculty  : Amanpreet Maam" << endl;
    cout << "9. Goal     : AI/ML Engineer" << endl;
    cout << "10. Thank you!" << endl;
    return 0;
}
```

Output:



Q3. Program to show Arithmetic Operators (+, -, *, /, %).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 20, b = 6;
    cout << "a + b = " << a + b << endl;
    cout << "a - b = " << a - b << endl;
    cout << "a * b = " << a * b << endl;
    cout << "a / b = " << a / b << endl;
    cout << "a % b = " << a % b << endl;
    return 0;
}
```

Output:



Q4. Program to show Assignment Operators (=, +=, -=, *=, /=, %=).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int x = 10;
    cout << "x = 10   -> " << x << endl;
    x += 5;  cout << "x += 5   -> " << x << endl;
    x -= 3;  cout << "x -= 3   -> " << x << endl;
    x *= 2;  cout << "x *= 2   -> " << x << endl;
    x /= 4;  cout << "x /= 4   -> " << x << endl;
    x %= 2;  cout << "x %= 2   -> " << x << endl;
    return 0;
}
```

Output:



Q5. Program to show Logical Operators (&&, ||, !).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 5, b = 10;
    cout << "a < b && b > 3 : " << (a < b && b > 3) << endl;
    cout << "a > b || b > 3 : " << (a > b || b > 3) << endl;
    cout << "!(a < b)       : " << !(a < b) << endl;
    return 0;
}
```

Output:



Q6. Program to show Ternary Operator (?:).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 15, b = 25;
    int big = (a > b) ? a : b;
    cout << "Bigger number = " << big << endl;
    string r = (b % 2 == 0) ? "Even" : "Odd";
    cout << b << " is " << r << endl;
    return 0;
}
```

Output:



Q7. Program to show Unary Operators (++, --, -).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 5;
    cout << "a++ : " << a++ << endl;
    cout << "a   : " << a << endl;
    cout << "++a : " << ++a << endl;
    cout << "a-- : " << a-- << endl;
    cout << "--a : " << --a << endl;
    cout << "-a  : " << -a << endl;
    return 0;
}
```

Output:



Q8. Program to show Relational Operators (==, !=, >, <, >=, <=).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 8, b = 12;
    cout << "a == b : " << (a == b) << endl;
    cout << "a != b : " << (a != b) << endl;
    cout << "a > b  : " << (a > b) << endl;
    cout << "a < b  : " << (a < b) << endl;
    cout << "a >= b : " << (a >= b) << endl;
    cout << "a <= b : " << (a <= b) << endl;
    return 0;
}
```

Output:



Q9. Program to show Bitwise Operators (&, |, ^, ~, <<, >>).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 12, b = 10;
    cout << "a & b  = " << (a & b) << endl;
    cout << "a | b  = " << (a | b) << endl;
    cout << "a ^ b  = " << (a ^ b) << endl;
    cout << "~a     = " << (~a) << endl;
    cout << "a << 1 = " << (a << 1) << endl;
    cout << "a >> 1 = " << (a >> 1) << endl;
    return 0;
}
```

Output:



Q10. Check whether a number is Positive, Negative or Zero.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int num = -7;
    if (num > 0)
        cout << num << " is Positive" << endl;
    else if (num < 0)
        cout << num << " is Negative" << endl;
    else
        cout << "Number is Zero" << endl;
    return 0;
}
```

Output:



Q11. Check whether a number is Even or Odd.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int num = 14;
    if (num % 2 == 0)
        cout << num << " is Even" << endl;
    else
        cout << num << " is Odd" << endl;
    return 0;
}
```

Output:



Q12. Find the largest of three numbers.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 45, b = 78, c = 32;
    if (a >= b && a >= c)
        cout << a << " is the largest" << endl;
    else if (b >= a && b >= c)
        cout << b << " is the largest" << endl;
    else
        cout << c << " is the largest" << endl;
    return 0;
}
```

Output:



Q13. Print the grade of a student according to marks.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int marks = 78;
    if (marks >= 90)      cout << "Grade A+" << endl;
    else if (marks >= 75) cout << "Grade A" << endl;
    else if (marks >= 60) cout << "Grade B" << endl;
    else if (marks >= 40) cout << "Grade C" << endl;
    else                  cout << "Fail" << endl;
    return 0;
}
```

Output:



Q14. Check whether a year is a Leap Year or not.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int year = 2024;
    if ((year % 4 == 0 && year % 100 != 0) || year % 400 == 0)
        cout << year << " is a Leap Year" << endl;
    else
        cout << year << " is not a Leap Year" << endl;
    return 0;
}
```

Output:



Q15. Find the type of a triangle from its three sides.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 5, b = 5, c = 8;
    if (a + b > c && b + c > a && a + c > b) {
        if (a == b && b == c)
            cout << "Equilateral Triangle" << endl;
        else if (a == b || b == c || a == c)
            cout << "Isosceles Triangle" << endl;
        else
            cout << "Scalene Triangle" << endl;
    } else
        cout << "Triangle is not possible" << endl;
    return 0;
}
```

Output:

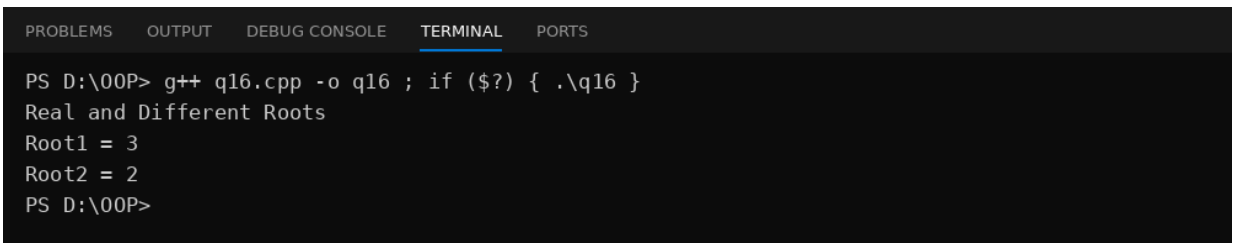


Q16. Find the nature and roots of a quadratic equation.

Program:

```
#include <iostream>
#include <cmath>
using namespace std;
int main() {
    int a = 1, b = -5, c = 6;
    int d = b * b - 4 * a * c;
    if (d > 0) {
        cout << "Real and Different Roots" << endl;
        cout << "Root1 = " << (-b + sqrt(d)) / (2 * a) << endl;
        cout << "Root2 = " << (-b - sqrt(d)) / (2 * a) << endl;
    } else if (d == 0)
        cout << "Real and Equal Root = " << -b / (2.0 * a) << endl;
    else
        cout << "Roots are Imaginary" << endl;
    return 0;
}
```

Output:



Q17. Check whether a given date is valid or not.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int d = 29, m = 2, y = 2023, maxDay = 31;
    if (m == 4 || m == 6 || m == 9 || m == 11)
        maxDay = 30;
    else if (m == 2) {
        if ((y % 4 == 0 && y % 100 != 0) || y % 400 == 0)
            maxDay = 29;
        else
            maxDay = 28;
    }
    if (m >= 1 && m <= 12 && d >= 1 && d <= maxDay)
        cout << "Valid Date" << endl;
    else
        cout << "Invalid Date" << endl;
    return 0;
}
```

Output:



Q18. Make a simple calculator using if-else ladder.

Program:

```
#include <iostream>
using namespace std;
int main() {
    double a = 12, b = 4;
    char op = '*';
    if (op == '+')      cout << a << " + " << b << " = " << a + b << endl;
    else if (op == '-') cout << a << " - " << b << " = " << a - b << endl;
    else if (op == '*') cout << a << " * " << b << " = " << a * b << endl;
    else if (op == '/' && b != 0) cout << a << " / " << b << " = " << a / b << endl;
    else cout << "Invalid operator" << endl;
    return 0;
}
```

Output:



Q19. Calculate an electricity bill by unit slabs.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int units = 250;
    double bill;
    if (units <= 100)
        bill = units * 5;
    else if (units <= 200)
        bill = 500 + (units - 100) * 7;
    else
        bill = 500 + 700 + (units - 200) * 10;
    cout << "Units = " << units << endl;
    cout << "Bill  = Rs. " << bill << endl;
    return 0;
}
```

Output:



Q20. Check whether a person is eligible to vote (if-else).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int age = 19;
    if (age >= 18)
        cout << "Eligible to vote" << endl;
    else
        cout << "Not eligible to vote" << endl;
    return 0;
}
```

Output:

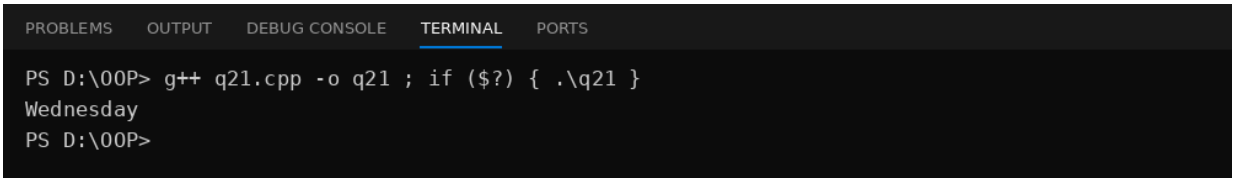


Q21. Print the day name from a day number (switch case).

Program:

```
#include <iostream>
using namespace std;
int main() {
    int day = 3;
    switch (day) {
        case 1: cout << "Monday" << endl;    break;
        case 2: cout << "Tuesday" << endl;   break;
        case 3: cout << "Wednesday" << endl; break;
        case 4: cout << "Thursday" << endl;  break;
        default: cout << "Invalid day" << endl;
    }
    return 0;
}
```

Output:

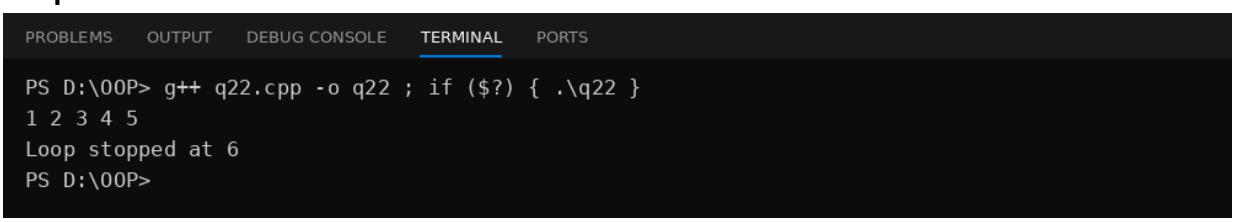


Q22. Print 1 to 10 but stop when the number is 6 (break).

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 6)
            break;
        cout << i << " ";
    }
    cout << endl << "Loop stopped at 6" << endl;
    return 0;
}
```

Output:

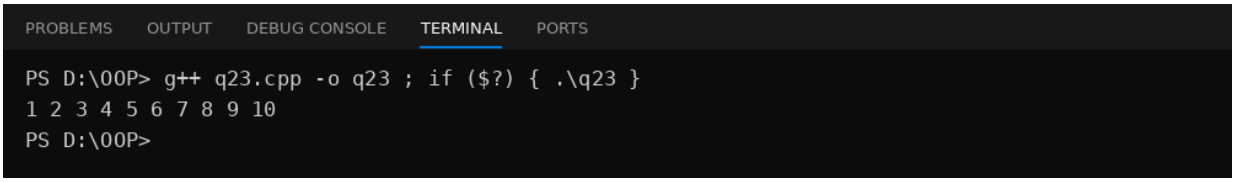


Q23. Print numbers from 1 to 10.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++)
        cout << i << " ";
    cout << endl;
    return 0;
}
```

Output:

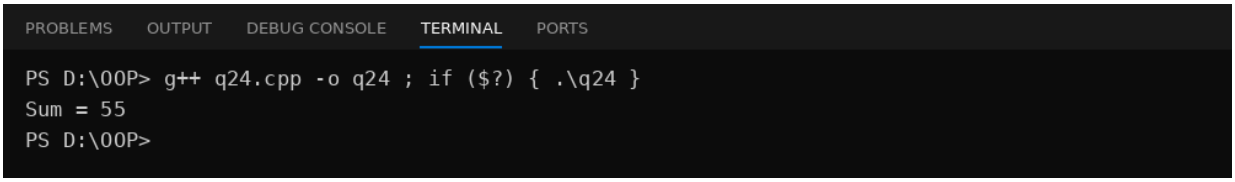


Q24. Find the sum of the first 10 natural numbers.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int sum = 0;
    for (int i = 1; i <= 10; i++)
        sum += i;
    cout << "Sum = " << sum << endl;
    return 0;
}
```

Output:

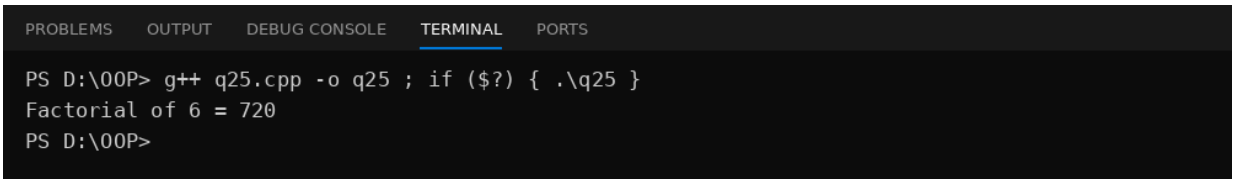


Q25. Find the factorial of a number.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int n = 6;
    long long fact = 1;
    for (int i = 1; i <= n; i++)
        fact *= i;
    cout << "Factorial of " << n << " = " << fact << endl;
    return 0;
}
```

Output:

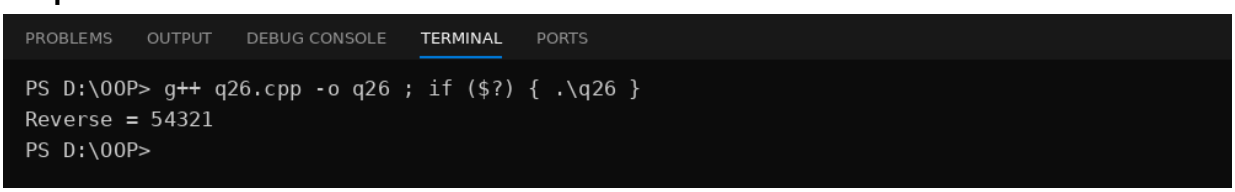


Q26. Reverse a number.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int n = 12345, rev = 0;
    while (n > 0) {
        rev = rev * 10 + n % 10;
        n = n / 10;
    }
    cout << "Reverse = " << rev << endl;
    return 0;
}
```

Output:



Q27. Print all prime numbers from 1 to 50.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int n = 2; n <= 50; n++) {
        bool prime = true;
        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0) {
                prime = false;
                break;
            }
        }
        if (prime)
            cout << n << " ";
    }
    cout << endl;
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS D:\OOP> g++ q27.cpp -o q27 ; if ($?) { .\q27 }
2 3 5 7 11 13 17 19 23 29 31 37 41 43 47
PS D:\OOP>
```

Q28. Print the table of 9 using for loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 10; i++)
        cout << "9 x " << i << " = " << 9 * i << endl;
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

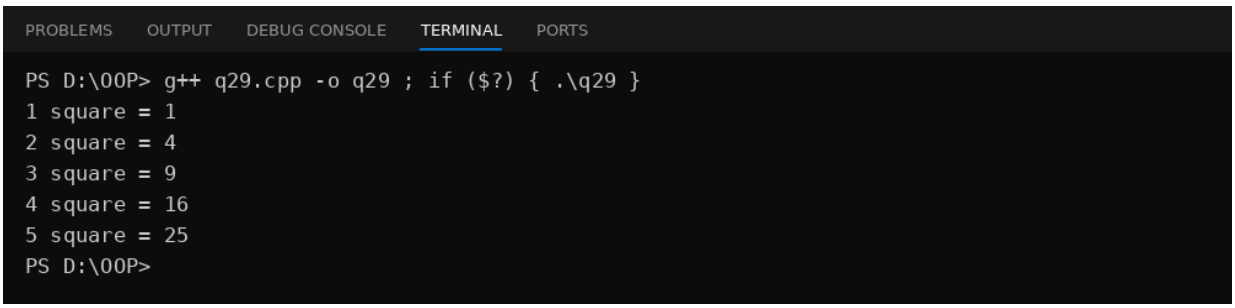
PS D:\OOP> g++ q28.cpp -o q28 ; if ($?) { .\q28 }
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
PS D:\OOP>
```

Q29. Print squares of 1 to 5 using for loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 5; i++)
        cout << i << " square = " << i * i << endl;
    return 0;
}
```

Output:

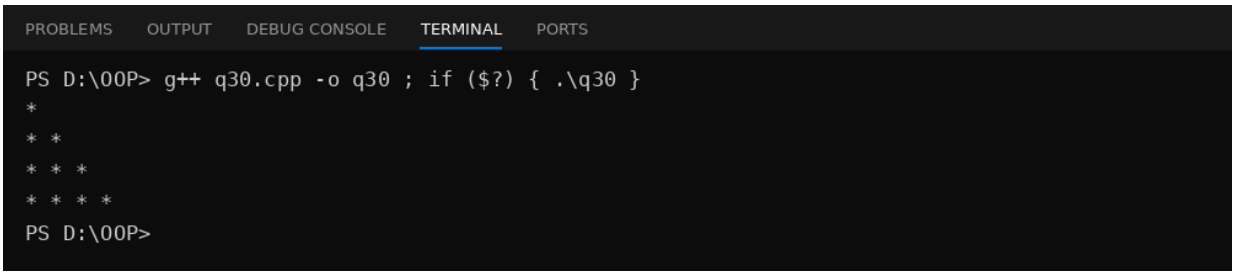


Q30. Print a star triangle pattern using nested for loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    for (int i = 1; i <= 4; i++) {
        for (int j = 1; j <= i; j++)
            cout << "* ";
        cout << endl;
    }
    return 0;
}
```

Output:



Q31. Print countdown from 5 to 1 using while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int i = 5;
    while (i >= 1) {
        cout << i << " ";
        i--;
    }
    cout << endl;
    return 0;
}
```

Output:

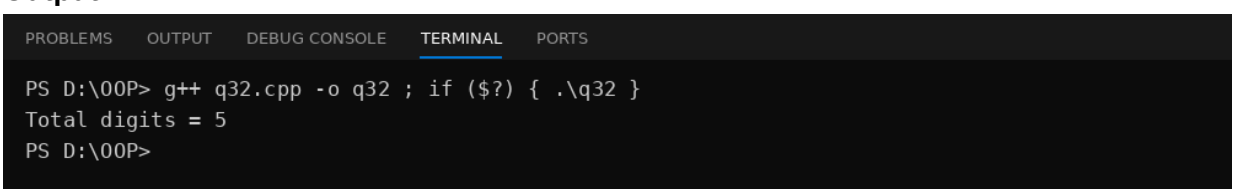


Q32. Count the digits of a number using while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int n = 98765, count = 0;
    while (n > 0) {
        count++;
        n = n / 10;
    }
    cout << "Total digits = " << count << endl;
    return 0;
}
```

Output:



Q33. Find the sum of digits of a number using while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int n = 4821, sum = 0;
    while (n > 0) {
        sum += n % 10;
        n = n / 10;
    }
    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

Output:

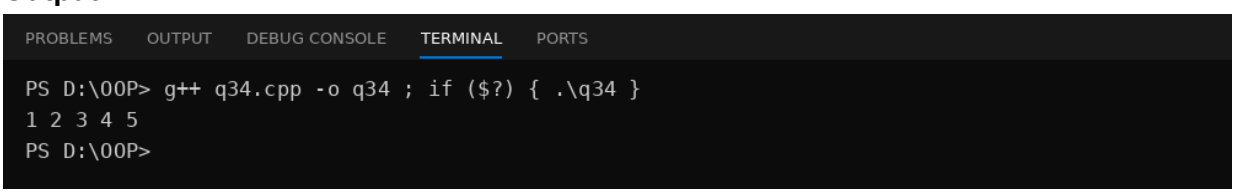


Q34. Print numbers from 1 to 5 using do-while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    do {
        cout << i << " ";
        i++;
    } while (i <= 5);
    cout << endl;
    return 0;
}
```

Output:

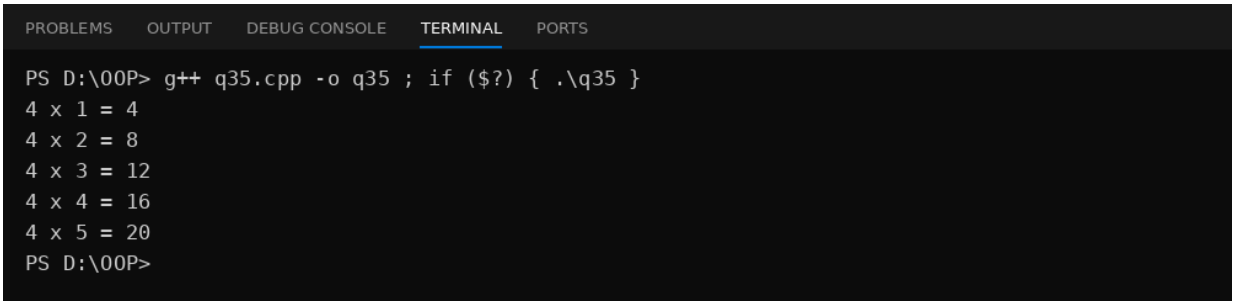


Q35. Print the table of 4 using do-while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int i = 1;
    do {
        cout << "4 x " << i << " = " << 4 * i << endl;
        i++;
    } while (i <= 5);
    return 0;
}
```

Output:

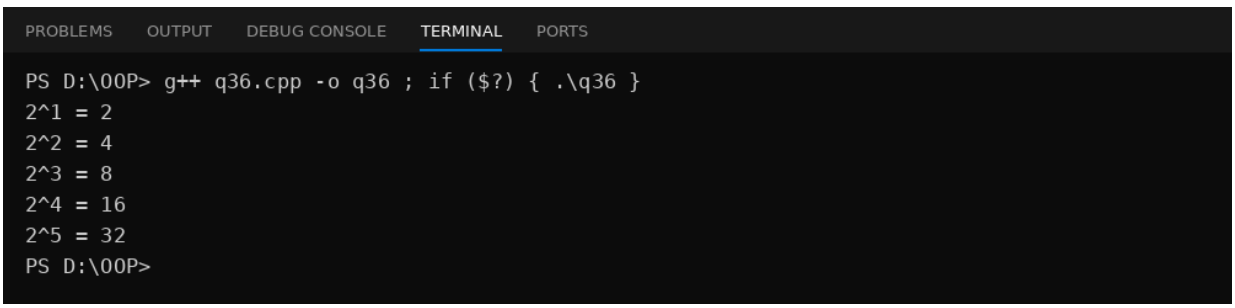


Q36. Print powers of 2 (2^1 to 2^5) using do-while loop.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int i = 1, p = 1;
    do {
        p = p * 2;
        cout << "2^" << i << " = " << p << endl;
        i++;
    } while (i <= 5);
    return 0;
}
```

Output:



Q37. 1D Array: print marks of 5 students and find the highest.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int marks[5] = {45, 78, 32, 90, 56};
    int big = marks[0];
    for (int i = 0; i < 5; i++) {
        cout << "marks[" << i << "] = " << marks[i] << endl;
        if (marks[i] > big)
            big = marks[i];
    }
    cout << "Highest marks = " << big << endl;
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS D:\OOP> g++ q37.cpp -o q37 ; if ($?) { .\q37 }
marks[0] = 45
marks[1] = 78
marks[2] = 32
marks[3] = 90
marks[4] = 56
Highest marks = 90
PS D:\OOP>
```

Q38. 2D Array: print a 2x3 matrix and the sum of its elements.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int m[2][3] = {{1, 2, 3}, {4, 5, 6}};
    int sum = 0;
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 3; j++) {
            cout << m[i][j] << " ";
            sum += m[i][j];
        }
        cout << endl;
    }
    cout << "Sum = " << sum << endl;
    return 0;
}
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS D:\OOP> g++ q38.cpp -o q38 ; if ($?) { .\q38 }
1 2 3
4 5 6
Sum = 21
PS D:\OOP>
```

Q39. Multi-dimensional (3D) Array: print a 2x2x2 array.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a[2][2][2] = {{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}};
    for (int i = 0; i < 2; i++) {
        cout << "Layer " << i + 1 << ":" << endl;
        for (int j = 0; j < 2; j++) {
            for (int k = 0; k < 2; k++)
                cout << a[i][j][k] << " ";
            cout << endl;
        }
    }
    return 0;
}
```

Output:

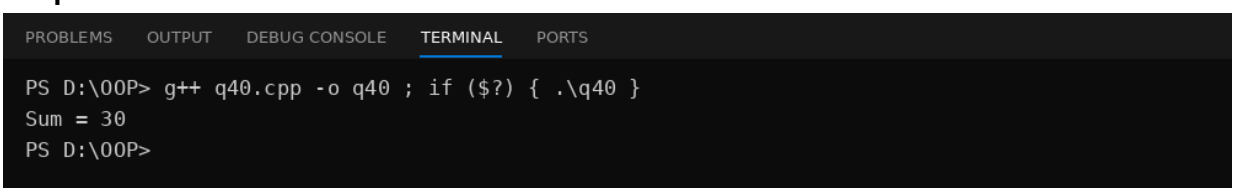


Q40. Function: add two numbers using a function.

Program:

```
#include <iostream>
using namespace std;
int add(int a, int b) {
    return a + b;
}
int main() {
    cout << "Sum = " << add(10, 20) << endl;
    return 0;
}
```

Output:

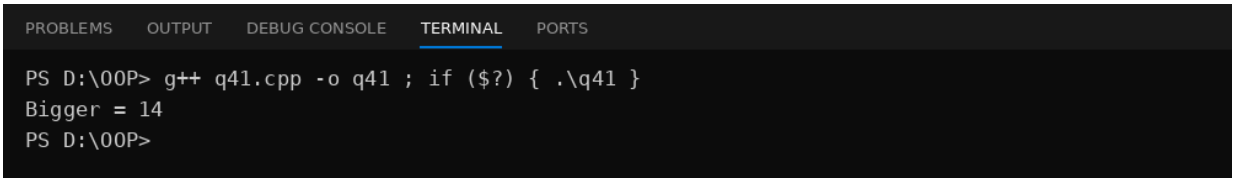


Q41. Function: find the bigger of two numbers.

Program:

```
#include <iostream>
using namespace std;
int bigger(int a, int b) {
    if (a > b)
        return a;
    return b;
}
int main() {
    cout << "Bigger = " << bigger(14, 9) << endl;
    return 0;
}
```

Output:

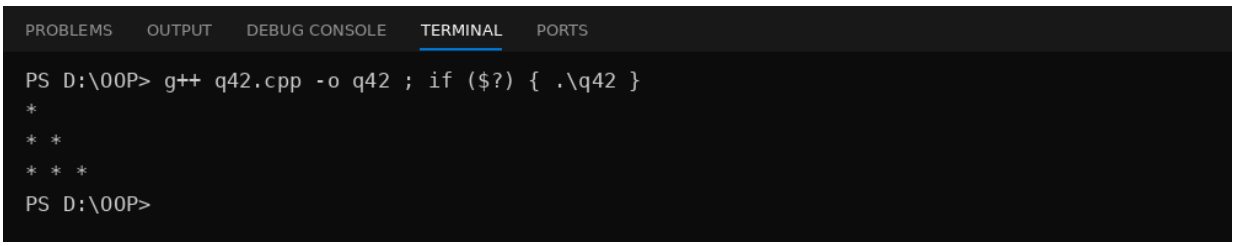


Q42. Function Type 1: No argument, no return value.

Program:

```
#include <iostream>
using namespace std;
void pattern() {
    for (int i = 1; i <= 3; i++) {
        for (int j = 1; j <= i; j++)
            cout << "* ";
        cout << endl;
    }
}
int main() {
    pattern();
    return 0;
}
```

Output:

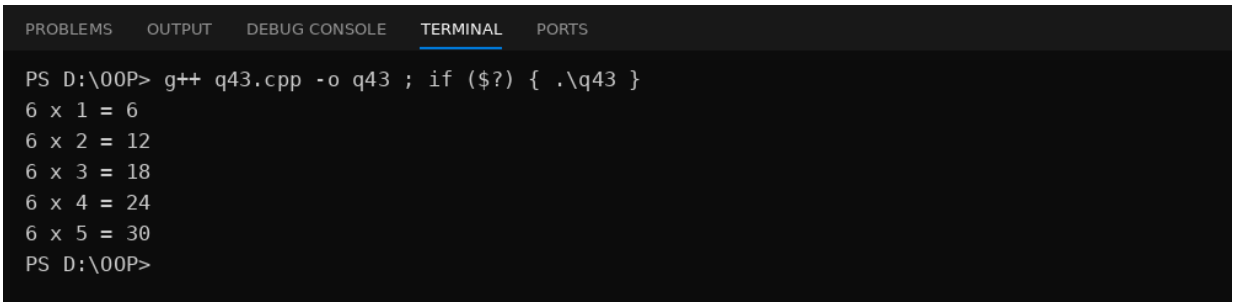


Q43. Function Type 2: With argument, no return value.

Program:

```
#include <iostream>
using namespace std;
void printTable(int n) {
    for (int i = 1; i <= 5; i++)
        cout << n << " x " << i << " = " << n * i << endl;
}
int main() {
    printTable(6);
    return 0;
}
```

Output:

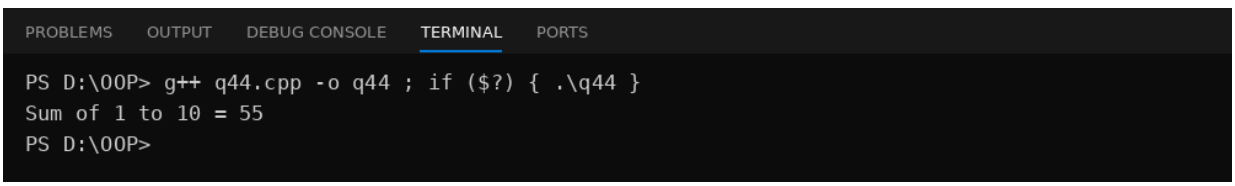


Q44. Function Type 3: No argument, with return value.

Program:

```
#include <iostream>
using namespace std;
int getSum() {
    int s = 0;
    for (int i = 1; i <= 10; i++)
        s += i;
    return s;
}
int main() {
    cout << "Sum of 1 to 10 = " << getSum() << endl;
    return 0;
}
```

Output:

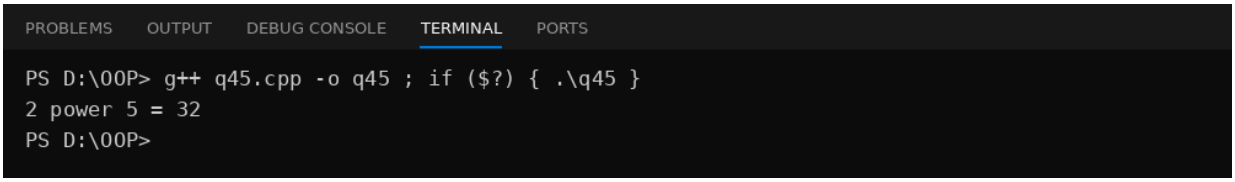


Q45. Function Type 4: With argument, with return value.

Program:

```
#include <iostream>
using namespace std;
int power(int base, int exp) {
    int r = 1;
    for (int i = 1; i <= exp; i++)
        r *= base;
    return r;
}
int main() {
    cout << "2 power 5 = " << power(2, 5) << endl;
    return 0;
}
```

Output:

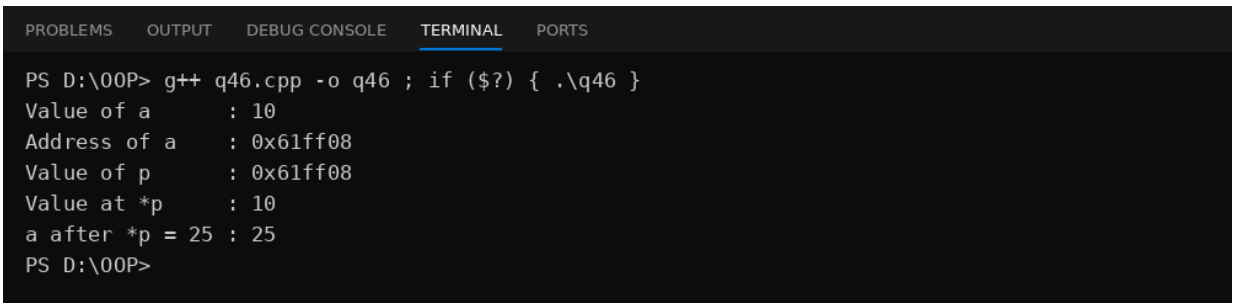


Q46. Pointer: show the basics of a pointer.

Program:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10;
    int *p = &a;
    cout << "Value of a      : " << a << endl;
    cout << "Address of a    : " << &a << endl;
    cout << "Value of p      : " << p << endl;
    cout << "Value at *p     : " << *p << endl;
    *p = 25;
    cout << "a after *p = 25 : " << a << endl;
    return 0;
}
```

Output:

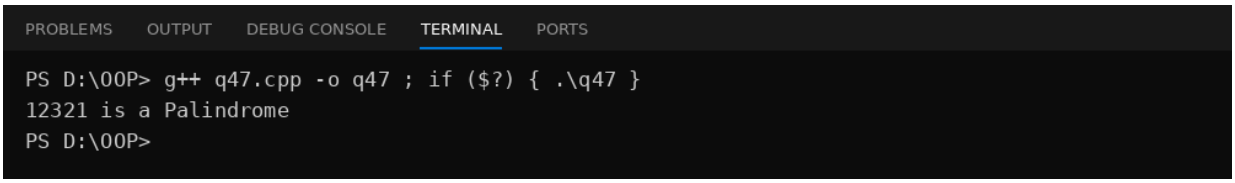


Q47. Function: check whether a number is a Palindrome.

Program:

```
#include <iostream>
using namespace std;
bool isPalindrome(int n) {
    int rev = 0, temp = n;
    while (temp > 0) {
        rev = rev * 10 + temp % 10;
        temp = temp / 10;
    }
    return rev == n;
}
int main() {
    int num = 12321;
    if (isPalindrome(num))
        cout << num << " is a Palindrome" << endl;
    else
        cout << num << " is not a Palindrome" << endl;
    return 0;
}
```

Output:



Q48. Function: print the Fibonacci series (first 10 terms).

Program:

```
#include <iostream>
using namespace std;
void fibonacci(int n) {
    int a = 0, b = 1;
    for (int i = 1; i <= n; i++) {
        cout << a << " ";
        int c = a + b;
        a = b;
        b = c;
    }
    cout << endl;
}
int main() {
    fibonacci(10);
    return 0;
}
```

Output:

